

Assignment 1

Friday, October 2, 2026 9:20 PM

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Q1 (20 marks):

- What is the amplitude, frequency, and period of the following signals?
- Draw each of the signals on a time-domain plot and a frequency-domain plot.
 - $s(t) = -3\sin(12t)$
 - $s(t) = 4\sin(3\pi t)$

a)

i) amplitude = 3

$$\text{frequency} = 2\pi f = 12 = \frac{12}{2\pi} = \frac{6}{\pi} = \boxed{1.91}$$

$$\text{period} = \frac{\pi}{6} = 0.524$$

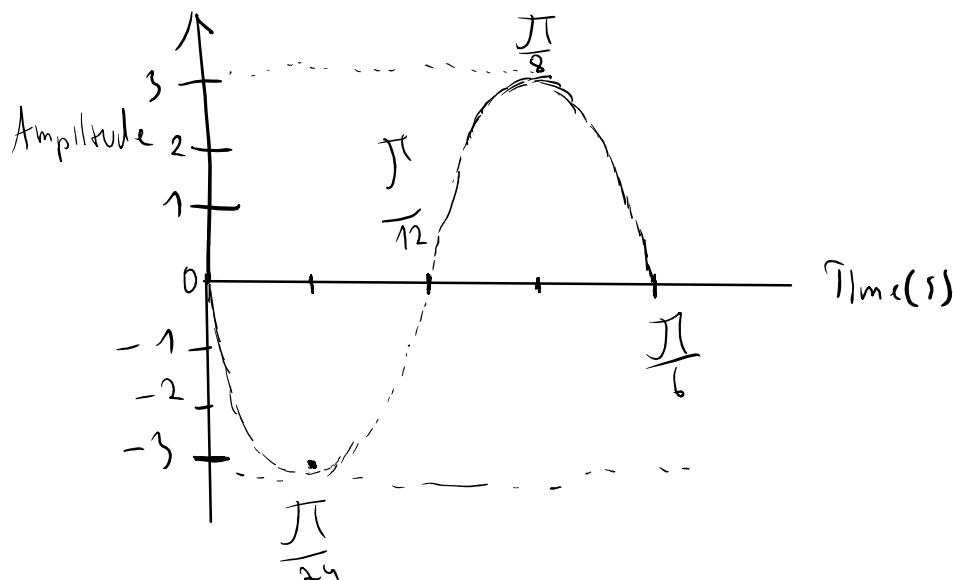
ii) amplitude = 4

$$\text{frequency} = 2\pi f = 3\pi = \frac{3\pi}{2\pi} = \frac{3}{2} = \boxed{1.5}$$

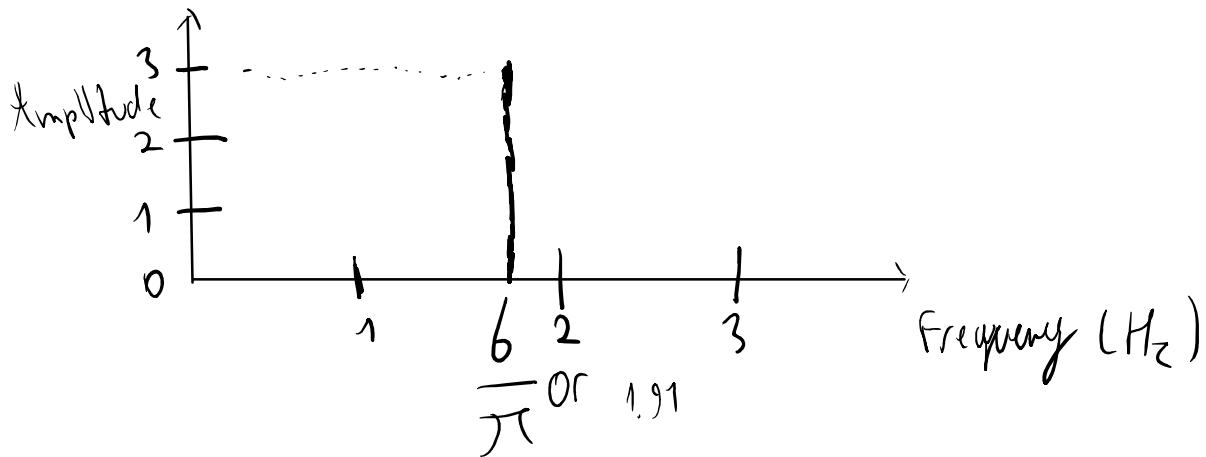
$$\text{period} = \frac{2}{3} = 0.667$$

$$b) i) 0, \frac{P}{4}, \frac{P}{2}, \frac{3P}{4}, P = 0, \frac{\pi}{24}, \frac{\pi}{12}, \frac{\pi}{8}, \frac{\pi}{6}$$

Time - Domain

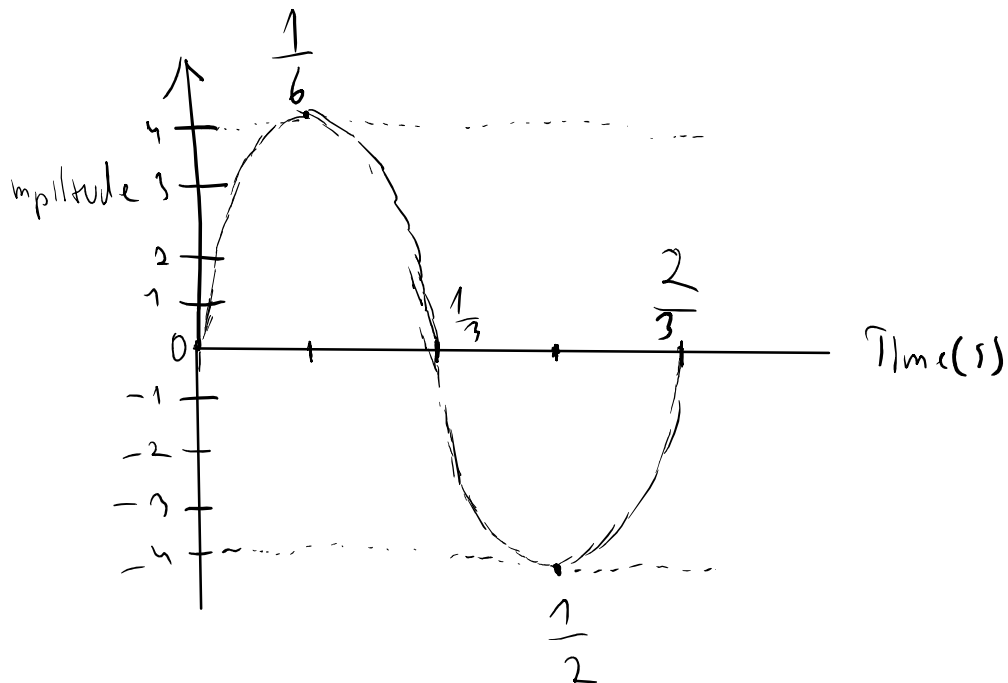


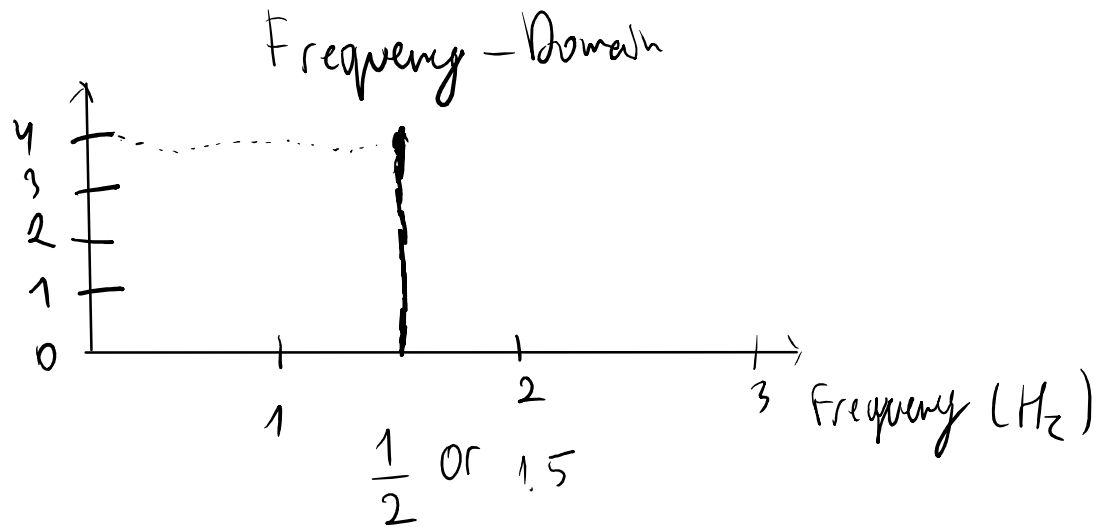
Frequency - Domain



ii) $0, \frac{p}{4}, \frac{p}{2}, \frac{3p}{4}, p = 0, \frac{1}{6}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}$

Time - Domain





Q2 (5 marks):

A signal has a wavelength of $1 \mu\text{m}$ in air. How far can the front of the wave travel during 1000 periods?

$$1 \times 10^{-6} \text{ m} \times 1 \times 10^3 = 10^{-3} \text{ m} \text{ or } \boxed{1 \text{ mm}}$$

Q3 (10 marks):

A computer monitor displays an image with a resolution of 1200×1000 pixels. Each pixel can represent 1024 different colors.

- How many bits are required to transmit the entire contents of the screen?
- Given a transmission rate of 1 Mbps (megabits per second), how long would it take to download 14 images?

a) $1200 \times 1000 = 1200000$ total pixels

$\log_2(1024) = 10$ bits per pixel

$1200000 \times 10 \text{ bits} = \boxed{12000000 \text{ bits}}$

b) $12000000 = 12 \text{ Mb}$

$\frac{1 \text{ sec}}{1 \text{ Mb}} \times \frac{12 \text{ Mb}}{1} = 12 \text{ sec per image}$

$12 \text{ sec} \times 14 = 168 \text{ sec or } \boxed{2.8 \text{ min}}$

Q4 (5 marks):

How can a composite signal be decomposed into its individual frequencies? Explain.

Periodic Composite signal decomposes into a series of simple sine waves with discrete frequencies.

Nonperiodic composite signal decomposes into a combination of an infinite number of simple sine waves with continuous frequencies across continuous range.

Q5 (5 marks):

Can we say whether a signal is periodic or nonperiodic by just looking at its frequency domain plot? How?

Periodic signal produces a spectrum with discrete frequencies. In frequency domain it's represented by separate individual spikes at specific frequencies.

Nonperiodic signal produces a spectrum with continuous frequencies.

In frequency domain it's represented by continuous curve spanning across a range of real-valued frequencies rather than spikes.

Q6 (5 marks):

A periodic composite signal contains frequencies from 10 to 30 kHz, each with an amplitude of 10 V. Draw the frequency spectrum.

